

TAHIYA ZAREEN HIYA

Software Developer | AI/ML

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SUMMARY

Computer Science and Engineering undergraduate with a strong interest in software development and artificial intelligence. Experienced in building web applications and working on practical AI/ML projects, with a focus on creating useful, reliable, and well-engineered software.

EDUCATION

University of Asia Pacific, Dhaka, Bangladesh	January 2023 – December 2026
B.Sc. in Computer Science and Engineering CGPA: 3.67 / 4.00	
Mohammadpur Preparatory School and College, Dhaka, Bangladesh	2021
Higher Secondary Certificate (HSC) GPA: 5.00 / 5.00	

AWARDS

Dean's Award, University of Asia Pacific	Spring 2024
Dean's Award, University of Asia Pacific	Fall 2024
Vice Chancellor's Award, University of Asia Pacific	Fall 2025

EXPERIENCE

Frontend Developer (Intern), Eutropia IT	Aug 2026 – Dec 2026
<i>Dhaka, Bangladesh</i>	
- Translate UI/UX design wireframes into high-quality, responsive code. - Collaborate with cross-functional teams to integrate APIs. - Ensure cross-browser compatibility and optimize application scalability. - Participate in code reviews to maintain high quality standards.	

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, TypeScript, HTML, CSS
Frontend Frameworks & Libraries: React, Next.js, Tailwind CSS, Vite, Framer Motion
Backend Frameworks & Runtimes: Django, FastAPI, Node.js
Mobile & Cross-Platform: Expo, Capacitor
Databases: PostgreSQL, MySQL
AI / Machine Learning: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, LLMs, NLP, RAG, Computer Vision, OCR, Federated Learning
Tools & Platforms: Git, GitHub, Linux, VS Code, Figma, Vercel, Render, Railway
Coding Agents: Cursor, Codex, Devin

SOFT SKILLS

Problem Solving, Self-Directed Learning, Attention to Detail, User-Centered Thinking, Technical Communication, Adaptability, Time Management

PROJECTS

EduGuard: Lightweight Local Offline Academic Assistant

2026

Undergraduate thesis project | Python, PyTorch, Hugging Face, llama.cpp | [Project document](#)

- Built an offline academic assistant combining LLM generation, Bloom's Taxonomy classification, retrieval, OCR-based document processing, and role-based student and teacher workflows.
- LoRa fine-tuned a Qwen2.5-0.5B model for six-level Bloom's Taxonomy classification and evaluated federated learning methods (FedAvg, FedProx).
- Developed training a Qwen2.5-1.5B for multitask generation pipeline Target level rewrite, Q&A, Summarization and deployed local GGUF inference with llama.cpp for CPU-based use.
- Implemented protected-content screening to support role aware privacy-maintaining academic assistance.

UAP CSE Assistant: RAG-Based University Website Assistant

2026

Web and Android application | Next.js, TypeScript, Capacitor, GitHub Actions | Live demo: uap-cse-bot.vercel.app

- Built a RAG assistant grounded in the UAP CSE website, with a crawler that collects HTML pages and links PDF/DOCX files into a retrieval corpus.
- Implemented semantic and lexical retrieval with metadata filtering, fuzzy person-name matching, and conversation-aware follow-up resolution.
- Automated knowledge-base updates with GitHub Actions: scheduled crawling, incremental embedding updates, and index validation.
- Delivered a Next.js web interface and an Android app via Capacitor.

ELIO: Rule-based Chatbot

2026

Web and Android application | React, Next.js, TypeScript, Capacitor | Live demo: elio-bot.vercel.app | [App download](#)

- Developed a rule-based chatbot with predefined conversation flows and animated interactions, packaged for Android with Capacitor.

Market Pulse: Mobile Application Prototype

2026

React Native, Expo, TypeScript, React Native | [App download](#)

- Developed a mobile prototype for browsing and filtering fictional disclosed-insider activity, with searchable screening, trade-detail views, and responsive layouts.

UAP CSE Department Website Redevelop

HTML, CSS, JavaScript, Django | Live site: cse.uap-bd.edu

2025

- Redesigned the department website with a focus on information structure, accessibility, and responsive presentation.

RESEARCH - Ongoing work

EduGuard: Federated Sub-Billion Bloom Classification and Multitask Academic Generation for Privacy-Aware University Assistance

Undergraduate Thesis, University of Asia Pacific

Topics: small language models, Bloom's Taxonomy classification, content cognitive level preserved question rewrite, retrieval-augmented generation, local inference, federated learning, OCR.